ADJETEI MARTEY

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PROFESSIONAL SUMMARY

Software engineer with an MSc in Artificial Intelligence (Distinction) building production ML systems and LLM-powered tools — from recommendation engines serving a live customer base to RAG pipelines, tool-calling agents, and computer vision models shipped as working demos.

TECHNICAL SKILLS

- **Programming Languages:** Python, TypeScript, C++, Java, SQL
- **ML/AI:** PyTorch, TensorFlow, Scikit-Learn, HuggingFace, Vercel AI SDK, RAG, agentic tool-calling, embeddings
- **Cloud & Infrastructure:** AWS, Google Cloud, Microsoft Azure, MongoDB, Docker, Kubernetes
- **AI Coding Agents:** Claude Code (daily use for building, refactoring and automating workflows)
- **Core Expertise:** Machine Learning, LLM Application Development, NLP, Database Optimization, API Development

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Nguvu Global Ltd — 2022 – Present

- Lead design and deployment of recommendation-system models for customer targeting on transactional data, serving a 10K-customer base — architecture now used as the foundation for the team's targeting work
- Built and shipped a prototype customer support chatbot powered by Anthropic's API — first GenAI integration into the product
- Reduced database query times by 40% by adding indexes to high-traffic queries
- Cut production bugs by 35% by introducing mandatory code review and regular change logging

Software Engineer | Nguvu Global Ltd — Sept 2019 – 2022

- Designed and shipped the initial recommendation-system model for customer targeting using transactional data, achieving a 30% increase in customer acquisition and 55% revenue growth in the first 3 months post-launch, confirmed via before/after sales comparison
- Built customer-facing platform features improving scalability for a 6-person engineering team supporting a growing customer base
- Provided IT support and network administration in the early stage of the team

PROJECTS

docuchat — RAG document Q&A | github.com/saxumatrum/docuchat

- Built a full RAG pipeline: overlapping-chunk splitting, batched embeddings (OpenAI text-embedding-3-small), in-memory cosine-similarity retrieval, and streamed LLM generation with inline source citations (Next.js, Vercel AI SDK)

TriagePilot — agentic ticket triage | github.com/saxumatrum/triagepilot

- Built a multi-step tool-calling agent that classifies support tickets, searches a knowledge base, and drafts replies, with a full inspectable trace of every tool call and result (Vercel AI SDK generateText + tools + stopWhen)

Extractly — document data extraction | github.com/saxumatrum/extractly

- Built a vision-LLM pipeline turning invoices, receipts, and resumes (image/PDF) into structured, schema-validated JSON via generateObject + Zod, with inline editing and CSV/JSON export

Chest X-Ray Multilabel Classification | github.com/saxumatrum/chest-xray-multilabel

- Fine-tuned DenseNet121 (CheXNet architecture) on NIH ChestX-ray14 for 14-way multilabel pathology classification; BCE loss, per-label AUROC evaluation

EDUCATION

University of Kent, Canterbury

Master of Science in Artificial Intelligence

- Graduated with Distinction (2023)
- Relevant Coursework: Artificial Intelligence, Cognitive Neural Networks, Deep Learning, Data Mining and Knowledge Discovery
- Research: Investigated genetic interactions influencing ageing under dietary restrictions using Balanced Random Forest Classification, XGBoost, and Random Forest models

University of Ghana, Legon

Bachelor of Science in Information Technology

- Second Class Honours (2021)
- Notable Research: COVID-19 Predictive Modelling using ARIMA and LSTM models
- Relevant Coursework: Object-Oriented Programming, Database Fundamentals, Cloud Computing, Network Architecture and Security